# C Programming Mini Project

**Project Title**: Console Based Snake Game using C

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## Abstract

The primary objective of the Snake Game project is to design and develop a console-based interactive application using the C programming language. The program creates a 20×20 grid-based game board in which the player controls a snake represented by the character 'O'. The main goal of the game is to navigate the snake across the board using keyboard inputs in order to consume the fruit represented by the character 'F'. Each time the snake successfully eats the fruit, the score increases, and a new fruit is generated at a random position on the board. The game continues until the snake collides with the boundary wall, at which point the game terminates.

This project aims to strengthen the understanding of fundamental C programming concepts such as loops, conditional statements, functions, and random number generation. It also focuses on improving logical thinking and problem-solving skills by implementing real-time movement control, collision detection, and score management mechanisms. Furthermore, the project provides hands-on experience in creating interactive console applications, thereby laying a strong foundation for advanced topics such as game development, graphics programming, and data structure implementation

# Introduction

The Snake Game is a simple console-based application developed using the C programming language.
It creates a 20×20 game board where the player controls a snake to eat fruit and increase the score.
The game ends when the snake hits the boundary wall.
This project demonstrates the use of loops, conditions, functions, and random number generation in C.

# Objectives

The main objective of this project is to design and develop a console-based Snake Game using the C programming language. This project aims to strengthen understanding of structured programming concepts such as loops, conditional statements, functions, and random number generation. It focuses on building logical thinking skills by implementing real-time movement control, collision detection, and score updating mechanisms within a simple game environment.

## Tools & Technology

Programming Language: C
Compiler: GCC / Turbo C / CodeBlocks
Platform: Windows / Linux

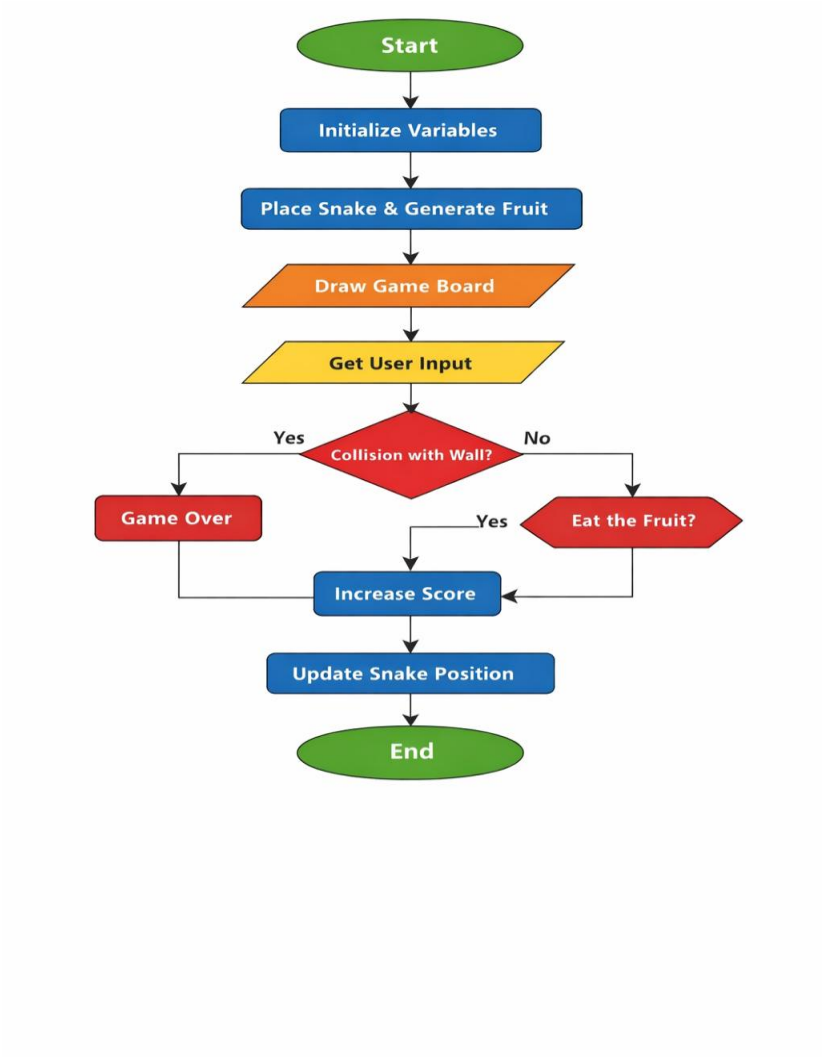
## System Requirements

Hardware: Basic computer with minimum 4GB RAM
Software: C Compiler and any IDE/Text Editor

## Methodology / Algorithm

1. Start the program.
2. Initialize variables such as snake position, fruit position, score, and game status.
3. Place the snake at the center of the 20×20 board.
4. Generate fruit at a random position within the board.
5. Display the game board with boundaries, snake (‘O’), and fruit (‘F’).
6. Read user input (W, A, S, D for movement and X to exit).
7. Update snake position based on the user’s input.
8. Check for collision with the boundary walls.
    - If collision occurs, set gameOver = 1.
9. Check if snake eats the fruit (snake position equals fruit position).
    - Increase score by 10.
    - Generate a new fruit at a random position.
10. Repeat steps 5–9 until the game is over.
11. Display final score.
12. End the program.

## FLOW CHART




## Program Code

```
#include <stdio.h>

#include <stdlib.h>

#include <time.h>

#define WIDTH 20

#define HEIGHT 20

int x, y, fruitX, fruitY, score;
```

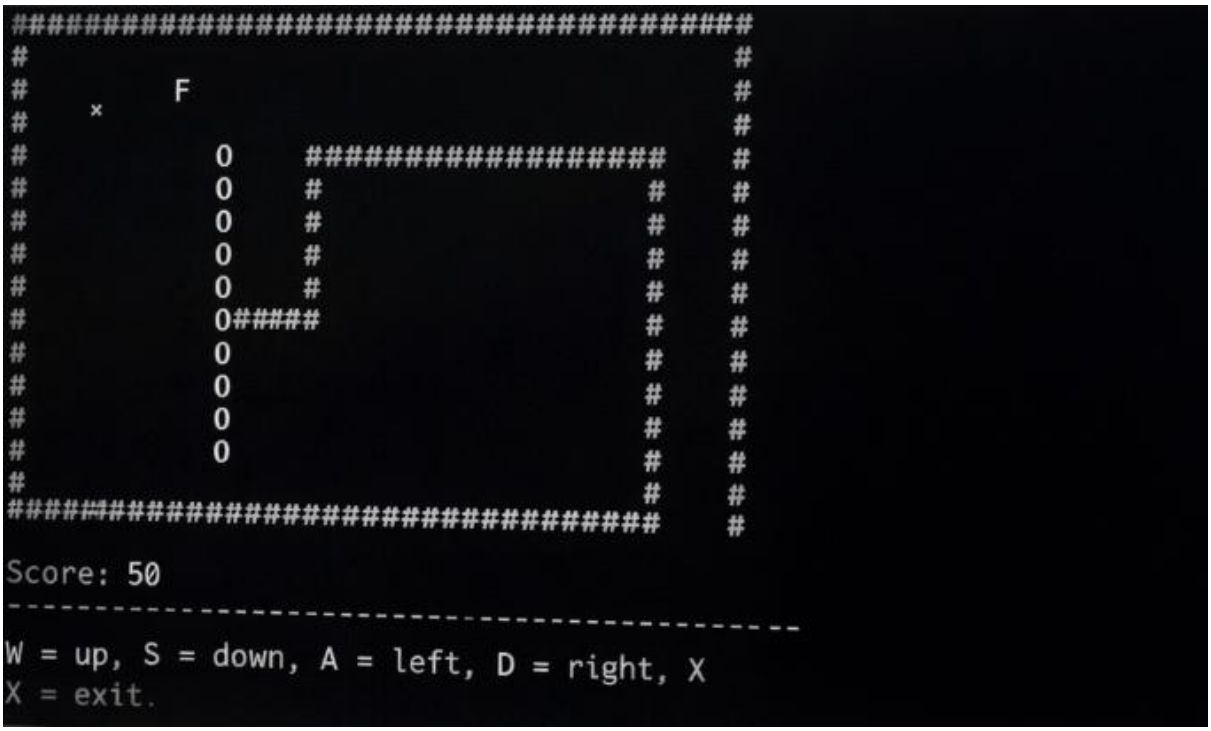
```
int gameOver;
void setup(){
    gameOver = 0;
    x = WIDTH / 2;
    y = HEIGHT / 2;
    srand(time(0));
    fruitX = rand() % WIDTH;
    fruitY = rand() % HEIGHT;
    score = 0;
}
void draw(){
    system("clear");   // use "cls" if Windows
    for (int i = 0; i < WIDTH + 2; i++)
        printf("#");
    printf("\n");
    for (int i = 0; i < HEIGHT; i++){
        for (int j = 0; j < WIDTH; j++){
            if (j == 0)
                printf("#")
if (i == y && j == x)
                printf("O");
```

```
            else if (i == fruitY && j == fruitX)
                printf("F");
            else
                printf(" ");
            if (j == WIDTH - 1)
                printf("#");
        }
        printf("\n");
    }
for (int i = 0; i < WIDTH + 2; i++)
        printf("#");

    printf("\nScore: %d\n", score);
    printf("Use W A S D to move. X to exit\n");
}
void input(){
    char ch;
    scanf(" %c", &ch);

    switch (ch){
        case 'a': x--; break;
```

```
        case 'd': x++; break;
        case 'w': y--; break;
        case 's': y++; break;
        case 'x': gameOver = 1; break;
    }
}
void logic(){
    if (x >= WIDTH || x < 0 || y >= HEIGHT || y < 0)
        gameOver = 1;
    if (x == fruitX && y == fruitY){
        score += 10;
        fruitX = rand() % WIDTH;
        fruitY = rand() % HEIGHT;
    }
}
int main(){
    setup();
    while (!gameOver){
        draw();
        input();
        logic();
```

```
    }
    printf("Game Over! Final Score: %d\n", score);
    return 0;
}
```

**Sample Output**




## Result

The program executed successfully and produced the expected output.

## Applications

- Used for academic learning and understanding basic C programming concepts
- Helps improve logical thinking and problem-solving skills
- Demonstrates practical use of loops, conditions, and functions
- Provides basic knowledge of game development concepts
- Useful for practicing structured programming techniques

- Can be extended to develop advanced graphical games

## 13. Conclusion

The Snake Game project is a simple console-based application developed using C.
It helps in understanding basic programming concepts like loops, conditions, and functions.
The project improves logical thinking and coding skills.
Overall, it is a good beginner-level project for learning C programming.

## 14. Future Enhancements

- Add snake tail
- Add levels and speed control
- Implement high score system
- Improve graphics

## 15. References


- C Programming Books
- Online Tutorials
- Class Notes